

Problem Statement

Objective

This assignment aims to give you a practical understanding of model selection in real-world scenarios. You will learn how to analyse and preprocess data, extract meaningful features, and apply various machine learning models to uncover patterns and insights.

Through model evaluation and optimisation techniques, you will gain the ability to compare different approaches and select the most effective one. This process will enhance your skills in building reliable predictive models, improving decision making, and ensuring model accuracy and generalisation in diverse applications.

Business Value

Developing a predictive model for cardiovascular disease risk helps simplify the early stages of patient assessment. Instead of relying solely on time-consuming consultations and manual checks, a screening model can highlight individuals who may need quicker attention.

In this setting, choosing the right model becomes important: different approaches offer different strengths in terms of accuracy, interpretability, and the ability to capture complex patterns in patient data. Selecting an appropriate model ensures that high-risk cases are identified reliably while still supporting clinical decision-making.

This allows healthcare teams to prioritise patients more effectively, make better use of limited consultation time, and improve the overall flow of care.

Dataset Overview

Context

The data contains records of various patients attended to by a healthcare provider. These patients came to the healthcare provider with complaints related to cardiovascular disease.

The data records the vital statistics of each patient and whether they had the disease or not. Some of the relevant features associated with these records are the patients' height, weight, age, blood pressure, and so on.

Content

The dataset is stored as a CSV file and contains a collection of patient health data, which includes demographic information such as age, gender, height, and weight, alongside physiological measurements like blood pressure.

The data also captures various lifestyle factors, including cholesterol and glucose levels, smoking habits, alcohol consumption, and physical activity.

Scoring and Penalty

- **Total Marks: 150** (100 for code notebook, and 50 for report document)
- **Extension and Penalty:** As given in your learner handbooks

Instructions

1. Each learner should attempt this assignment individually.
2. Programming Language: Python
3. You will be provided with the dataset and a starter notebook. You have to perform all the tasks in the starter notebook only.
4. It is very important that you do not change any headings, subheadings, questions or tasks in your notebook as it can cause problems with grading.
5. The data will have inconsistencies and outliers please handle them as per your understanding and mention them in your report.
6. You are encouraged to search the web and consult AI tools for conceptual understanding. However, using plagiarised or AI-generated code is strictly prohibited and strongly discouraged.
7. Submitting plagiarised and AI-generated code or reports will result in significant penalties to your scores.

Submission Guidelines

1. To submit your solution, upload your solution files in the submission field.
2. You are required to upload your 2 solution files:
 - an **Interactive Python Notebook** (.ipynb) that contains your code, and
 - a **Report Document** (.pdf) that presents your visualisations, analysis, results, insights, and outcomes
3. Please note that the solution notebook should only be generated from the starter files provided to you.
4. The submitted Jupyter notebook, report and PPT should contain your name and the assignment title in the format: "CardioRisk_Prediction_<your_name>".
5. Mention all assumptions, if made, in the report.
6. The report should include the overall approach of the assignment, covering the problem statement, methodology, techniques used, and key insights and should present answers to the questions asked in the assignment, using visualisations to support your responses.

Results Expected from Learners

Briefly outline the problem statement and approach in the report document. Use visualisations in the PPT to support your answers for all questions in the assignment.

In the starter notebook, you will find headings, subheadings, and checkpoints stating the tasks you need to perform. The marks associated with each checkpoint will also be mentioned in the notebook. Keep in mind not to edit the cells with marking schemes and questions.

You can find a brief description of the tasks below.

1. **Data Understanding [2 marks]**
 - (a) Remove irrelevant columns [2 marks]
2. **Data Cleaning [8 marks]**
 - (a) Identify and handle redundant or invalid/illogical physiological values [6 marks]
 - (b) Fix data types [2 marks]
3. **EDA [27 marks]**
 - (a) Perform univariate analysis [12 marks]
 - (b) Perform correlation analysis [5 marks]
 - (c) Perform bivariate analysis [10 marks]
4. **Train-Test Split [5 marks]**
 - (a) Define feature and target(s) variables [2 marks]
 - (b) Split data into training and validation sets [3 marks]
5. **Feature Engineering [18 marks]**
 - (a) New feature creation [6 marks]
 - (b) Combine values in categorical columns [4 marks]
 - (c) Dummy variable creation [5 marks]
 - (d) Feature rescaling [3 marks]
6. **Model Building [38 marks]**
 - (a) Build linear SVM and find optimal cutoff [18 marks]
 - (b) Build simple decision tree and find optimal cutoff [12 marks]
 - (c) Hyperparameter tuning [8 marks]
7. **Predictions and model evaluation [2 marks]**
 - (a) Make final predictions and evaluate [2 marks]
8. **Report [50 marks]**
 - (a) Exploratory Data Analysis and Feature Engineering [15 marks]
 - (b) Model Building and Hyperparameter Tuning [15 marks]
 - (c) Model cards, final insights and conclusions [20 marks]

Evaluation Rubrics

The following criteria will be used to evaluate your solutions.

Table 1: Rubrics

Criteria	Meets expectations	Does not meet expectations
Data Understanding	Identifier type columns were correctly handled.	Identifier type columns were handled incorrectly or were not handled at all.
Data Cleaning	- Data inconsistencies and errors were correctly identified and handled. - DataTypes have been corrected for columns with incorrect DataTypes.	- Data inconsistencies and errors were not identified and handled or were handled incorrectly. - DataTypes have not been corrected or have been incorrectly assigned for columns with incorrect DataTypes.
EDA	- Performed univariate analysis by visualising the distribution of all columns. - Performed correlation analysis by visualising a heatmap of the correlation matrix. - Perform bivariate analysis by visualising the relationship between categorical columns and the target variable.	- Failed to plot the distributions of numerical columns or created incomplete or inaccurate plots without meaningful insights. - Incorrectly visualised the heatmap or failed to interpret correlations. - Failed to visualise the influence of categorical variables on the target variable. Provided incomplete or unclear visualisations and insights.
Train-Test Split	- Feature and target variables are defined correctly. - Data is split into training and test sets maintaining the ratio of 70:30.	- Feature and target variables are not defined or defined incorrectly. - Data splitting is implemented incorrectly or not implemented at all.
Feature Engineering	- Created new feature(s) from existing features to enhance the model's ability to capture patterns. - Identified and combined low frequency values in categorical columns to reduce sparsity and improve model generalisation. - Created dummy variables for independent columns in both training and validation sets. - Applied feature scaling to numerical columns effectively by	- Failed to create new feature(s) from existing features to enhance the model's ability to capture patterns. - Failed to identify and combine low frequency values in categorical columns to reduce sparsity and improve model generalisation. - Failed to identify or create appropriate dummy variables for independent columns in both training and validation sets.

	scaling the features.	4. Failed to apply or incorrectly performed feature scaling, leading to inconsistent data ranges.
Model Building	1. Built the required models using the selected features, made predictions, and assessed model performance correctly. 2. Found the optimal cutoff for the models by plotting the ROC curve, visualising trade-offs between sensitivity and specificity, precision and recall, and evaluated the final prediction with optimal cutoff. 3. Tuned the decision tree model using appropriate technique, optimised hyperparameters, made predictions, and assessed performance with relevant metrics.	1. Failed to correctly build the models using selected features and evaluated or interpreted the performance metrics incorrectly. 2. Failed to identify the optimal cutoff for or omitted key evaluations like plotting curves and calculated necessary performance metrics. 3. Failed to tune the decision tree model effectively, did not optimise hyperparameters, or misinterpreted the performance evaluation.
Prediction and Model Evaluation	1. Predictions were made on validation data using the selected relevant features and hyperparameters. 2. Predictions were made on training and testing data using the models. 3. Evaluated the performance of all models using the appropriate evaluation metrics.	1. Failed to select relevant features and hyperparameters or incorrectly made predictions. 2. Incorrectly made predictions on testing or training data using the models. 3. Failed to evaluate the performance of models using the correct evaluation metrics or interpreted the results inaccurately.
Report and Recommendations	1. The report has a clear structure, is not too long, and explains the most important results concisely in simple language. 2. The recommendations to solve the problem are realistic, actionable and coherent with the analysis. 3. The report includes visualisations and insights derived from them. 4. If any assumptions are made, they are stated clearly.	1. The report lacks structure, is too long or does not put emphasis on the important observations. The language used is complicated for leaders to understand. 2. The recommendations to solve the problem are either unrealistic, non-actionable or incoherent with the analysis. 3. The report is missing visualisations or fails to provide meaningful insights. 4. Assumptions made, if any, are not stated clearly.

Conciseness and readability of the code	1. The code is concise and syntactically correct. Wherever appropriate, built-in functions and standard libraries are used instead of writing long code (if-else statements, loops, etc.). 2. Custom functions are used to perform repetitive tasks. 3. The code is readable with appropriately named variables and detailed comments are written wherever necessary.	1. Long and complex code is used instead of shorter built-in functions. 2. Custom functions are not used to perform repetitive tasks resulting in the same piece of code being repeated multiple times. 3. Code readability is poor because of vaguely named variables or lack of comments wherever necessary.
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